Why Sybil (sybild) CANNOT run as a remote Active Directory client

sybild impersonates by FORGING a TGT signed with the realm krbtgt key, which it reads from the KDC database via kadm5 — AD exposes no such interface

WINDOWS • Active Directory

Active Directory Domain Controller — KDC

- issues tickets via the Kerberos protocol, but…
- exposes NO kadm5 / KDB key interface to clients
- the krbtgt key never leaves the domain controller
- pulling principal keys out of AD = DCSync
  = full domain compromise (an attack, not an API)
- S4U is supported — but that is a CLIENT protocol,
  not the key-level DB access sybild requires

sybild asks for the krbtgt key from the KDC DB

AD will never export it
→ sybild cannot operate against AD

LINUX • sybild (NVIDIA/sybil)

sybild — impersonation mechanism

- `sybil kinit user@REALM` always FORGES a TGT
  krb::Credentials::forge(user, “krbtgt/REALM”) [lib.rs:266]
- signs it with the realm krbtgt key, which it reads
  from the KDC DB: kadm5_get_principal_keys()
  [krbutil.c:159] — daemon runs kadm5_init() [:42]
- no S4U-only mode — forging is unconditional
- ⇒ needs KDB access + co-location with an
  MIT / FreeIPA KDC

What actually works against AD → orthrus

orthrus is a Kerberos S4U *client* of AD: it holds a sybil-style SERVICE keytab and calls S4U2Self / S4U2Proxy over the wire — AD issues every ticket and authorizes delegation via msDS-AllowedToDelegateTo. No KDB access, no key reading, no co-location with the KDC.
Same goal as Sybil (impersonate batch users), DIFFERENT and AD-supported mechanism (constrained delegation, not forging). See docs/orthrus-architecture.

— blocked: key-level DB access AD does not provide

Verified in NVIDIA/sybil source — lib.rs:266, krbutil.c:42/159/214-286